

Name _____ Date _____

Probability Quiz



I "mustache" you to follow the directions below and do your best!

Vocabulary: Match each vocabulary word to its definition by writing the corresponding letter on the line next to the word.

1. _____ Probability
2. _____ Even Chance
3. _____ Certain
4. _____ Impossible
5. _____ Likely
6. _____ Unlikely
7. _____ Outcome
8. _____ Tree Diagram

- A. A possible result of an event, game, or experiment.
- B. A display to show all possible outcomes
- C. The chance of an event happening.
- D. An event cannot happen, and the probability is zero.
- E. An event will absolutely happen, and the probability is one.
- F. The chance of each event happening is equal.
- G. An event ~~probably~~ will NOT happen.
- H. An event ~~probably~~ WILL happen.

Probability: Solve the following problems. Fractions should be expressed in simplest form.

9) What is the probability of rolling a number equal to or greater than 7 on a ten-sided die (numbered 1-10)?

Final Answer: _____

10) On a multiple choice test, each question has four possible answer choices (A,B,C,D). On one of the questions, you randomly pick an answer because you don't know it. What is the probability you will get it correct?

Final Answer: _____

11) Seven boys and thirteen girls have all written their names on a piece of paper and added it to a bowl. You need to pick one name at random, what is the probability you will pick a boy's name?

Final Answer: _____

12) Your friend wants you to pick one number between 1 and 30. What is the probability you will pick a multiple of five?

Final Answer: _____

13) The letters of the word STUDENT are each written on a separate card and placed face down on a table. What is the probability, if you select a card at random, that you will choose a consonant?

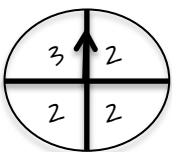
Final Answer: _____

14) Circle the best answer to complete the sentence.

An event with a probability of $\frac{1}{10}$ is _____.

- A) is certain to happen C) is likely to happen
B) is impossible D) is unlikely to happen

15) If you spin the spinner below once, what is the probability you will land on an even number?



Final Answer: _____

16) At the local bakery, you can pick one free cupcake, but you cannot look at which one you are picking. If there are 5 red velvet cupcakes, 6 chocolate cupcakes, 3 carrot cupcakes, and 12 vanilla cupcakes, what is the probability you will choose a chocolate OR vanilla cupcake?

Final Answer: _____

Probability Quiz continued...

Name _____

17) In a gumball machine, there are 6 red gumballs, 4 blue gumballs, and three green gumballs. What is the probability you will select a yellow gumball?

Final Answer: _____

18) Circle the best answer to complete the sentence.

You won a free cookie with your lunch, but the cookie will be selected at random. There are 10 chocolate chip cookies, 4 oatmeal raisin, and 6 sugar cookies. The probability that you will choose a chocolate chip cookie _____.

A) is certain.

B) is unlikely.

C) is impossible.

D) is an even chance.

19) Your school is casting for the new school play. Two students are trying out to be the director of the play: Mandy and Ryan. There are three people trying out for the lead role: Amanda, Betsy, and Rachel. There are also three people trying out for the supporting role: Jake, Tom, and Brian. Create a tree diagram to show all the possibilities for casting the director, lead role, and supporting role. The labels have been done for you. BE SURE TO ANSWER THE QUESTIONS BELOW!!!

Director

Lead Role

Supporting Role

Possible Outcomes

20) How many possible outcomes are there for casting the director, lead role, and supporting role in the school play? _____

*Bonus: What is the probability Mandy will be the director, Betsy will play the lead role, and Brian will play the supporting role? _____

Probability Quiz

Name ANSWER KEY

Date _____



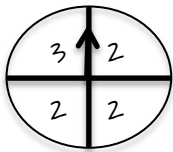
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Vocabulary: Match each vocabulary word to its definition by writing the corresponding letter on the line next to the word.

1. C Probability
2. F Even Chance
3. E Certain
4. D Impossible
5. H Likely
6. G Unlikely
7. A Outcome
8. B Tree Diagram

- A. A possible result of an event, game, or experiment.
- B. A display to show all possible outcomes
- C. The chance of an event happening.
- D. An event cannot happen, and the probability is zero.
- E. An event will absolutely happen, and the probability is one.
- F. The chance of each event happening is equal.
- G. An event ~~probably~~ will NOT happen
- H. An event ~~probably~~ WILL happen

Probability: Solve the following problems Fractions should be expressed in simplest form

9) What is the probability of rolling a number equal to or greater than 7 on a ten-sided die (numbered 1-10)? Final Answer: <u>$\frac{4}{10}$ or $\frac{2}{5}$ (decimal: 0.4)</u>	10) On a multiple choice test, each question has four possible answer choices (A,B,C,D). On one of the questions, you randomly pick an answer because you don't know it. What is the probability you will get it correct? Final Answer: <u>$\frac{1}{4}$ (decimal: 0.25)</u>
11) Seven boys and thirteen girls have all written their names on a piece of paper and added it to a bowl. You need to pick one name at random, what is the probability you will pick a boy's name? Final Answer: <u>$\frac{7}{20}$ (decimal: 0.35)</u>	12) Your friend wants you to pick one number between 1 and 30. What is the probability you will pick a multiple of five? Final Answer: <u>$\frac{6}{30}$ or $\frac{1}{5}$ (decimal: 0.2)</u>
13) The letters of the word STUDENT are each written on a separate card and placed face down on a table. What is the probability, if you select a card at random, that you will choose a consonant? Final Answer: <u>$\frac{5}{7}$ (decimal: 0.71)</u>	14) Circle the best answer to complete the sentence. An event with a probability of $\frac{1}{10}$ is _____. A) is certain to happen C) is likely to happen B) is impossible <u>D) is unlikely to happen</u>
15) If you spin the spinner below once, what is the probability you will land on an even number?  Final Answer: <u>$\frac{3}{4}$ (decimal: 0.75)</u>	16) At the local bakery, you can pick one free cupcake, but you cannot look at which one you are picking. If there are 5 red velvet cupcakes, 6 chocolate cupcakes, 3 carrot cupcakes, and 12 vanilla cupcakes, what is the probability you will choose a chocolate OR vanilla cupcake? Final Answer: <u>$\frac{18}{26}$ or $\frac{9}{13}$ (decimal: 0.69)</u>

Probability Quiz continued...ANSWER KEY

Name _____

17) In a gumball machine, there are 6 red gumballs, 4 blue gumballs, and three green gumballs. What is the probability you will select a yellow gumball?

Final Answer: Zero (decimal: 0) *impossible accepted

18) Circle the best answer to complete the sentence.

You won a free cookie with your lunch, but the cookie will be selected at random. There are 10 chocolate chip cookies, 4 oatmeal raisin, and 6 sugar cookies. The probability that you will choose a chocolate chip cookie _____.

A) is certain.
B) is unlikely.

C) is impossible.
D) is an even chance.

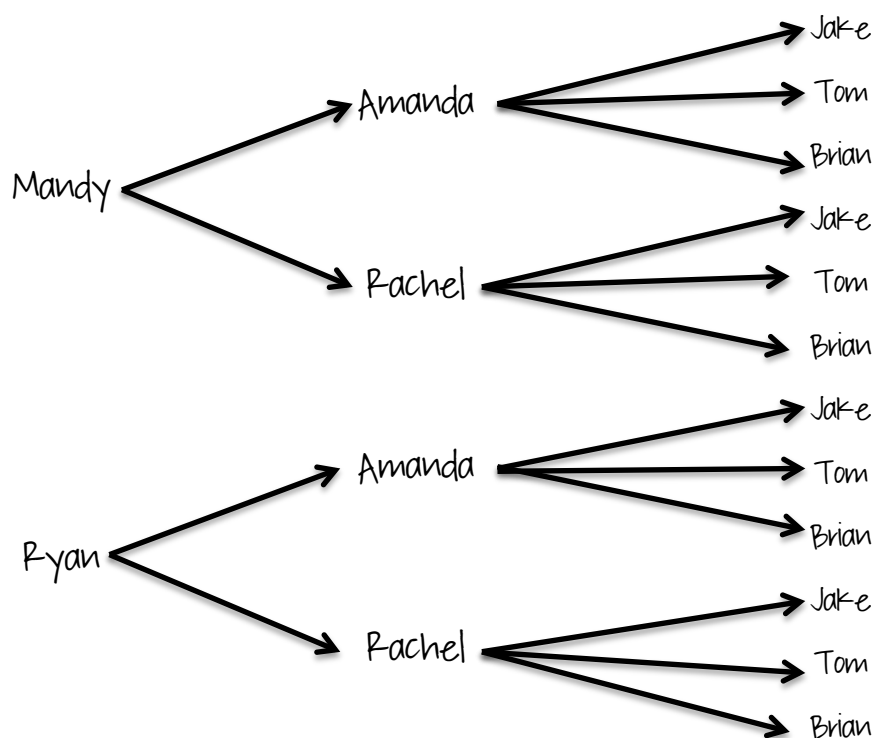
19) Your school is casting for the new school play. Two students are trying out to be the director of the play: Mandy and Ryan. There are two people trying out for the lead role: Amanda and Rachel. There are also three people trying out for the supporting role: Jake, Tom, and Brian. Create a tree diagram to show all the possibilities for casting the director, lead role, and supporting role. The labels have been done for you. BE SURE TO ANSWER THE QUESTIONS BELOW!!!

Director

Lead Role

Supporting Role

Possible Outcomes



Mandy, Amanda, Jake
Mandy, Amanda, Tom
Mandy, Amanda, Brian
Mandy, Rachel, Jake
Mandy, Rachel, Tom
Mandy, Rachel, Brian
Ryan, Amanda, Jake
Ryan, Amanda, Tom
Ryan, Amanda, Brian
Ryan, Rachel, Jake
Ryan, Rachel, Tom
Ryan, Rachel, Brian

20) How many possible outcomes are there for casting the director, lead role, and supporting role in the school play? 12 possible combinations

*Bonus: What is the probability Mandy will be the director, Betsy will play the lead role, and Brian will play the supporting role? One-twelfth or 1/12 (decimal: 0.08)